

SAMA: Open-Source Multi-Objective Optimization Platform for Hybrid Renewable Energy Microgrid Design

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Software

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Summary

Solar Alone Multi-Objective Advisor (SAMA) is an open-source platform for the planning, design, and operation optimization of hybrid renewable energy microgrids. Given location-specific electrical and thermal load profiles, meteorological data, and utility billing structures, SAMA finds the optimal combination of user-selected system components, including solar photovoltaic (PV) capacity, wind turbines, battery storage, backup generator, and inverter rating, that minimizes the lifecycle net present cost (NPC) of the system. Users choose which components to include in the optimization (e.g., PV-only, PV with battery storage, or full PV-battery-generator hybrid), and may optionally include a second objective, e.g., leveled emissions (LEM), to jointly optimize cost and environmental impact.

The platform simulates all hours of a representative year through an hourly economic dispatch engine, supports electric vehicles (EVs) with vehicle-to-home (V2H) and vehicle-to-grid (V2G) capabilities (user's choice), integrates air-source heat pump models (user's choice), and handles eight distinct electricity tariff structures including time-of-use (TOU) and tiered pricing. Four metaheuristic algorithms are available for the optimization: Particle Swarm Optimization (PSO), Advanced Differential Evolution (ADE), Artificial Bee Colony (ABC), and Grey Wolf Optimizer (GWO). SAMA is distributed as a pip-installable Python package, is available for direct download and execution from the GitHub repository ([Sadat, 2023](#)), and has been validated against HOMER Pro across multiple U.S. climate zones ([Sadat et al., 2023](#)).

Statement of Need

The global deployment of solar photovoltaic (PV) systems and hybrid renewable energy microgrids is accelerating rapidly, yet the software tools needed to optimally size and economically evaluate these systems remain inaccessible to much of the world ([Sadat et al., 2023](#)). Designing a hybrid system requires simultaneous optimization over competing objectives (minimizing lifecycle cost, meeting reliability constraints, maximizing renewable penetration) while accounting for location-specific meteorology, load profiles, and utility billing structures. Without appropriate software, practitioners resort to oversized or undersized systems, underperforming designs, or abandon renewable deployment entirely.

The dominant tool for microgrid planning and design, HOMER Pro, was originally developed as open-source software by the U.S. National Renewable Energy Laboratory (NREL) but has transitioned to a closed-source, for-profit model ([HOMER Energy, 2024b](#)). Licensing costs range from \$187.50 to \$568.50 per month for standard users, placing it out of reach for low-resource laboratories, researchers in developing regions, small municipalities, NGOs, and independent practitioners. Beyond cost, its closed-source nature prevents researchers and

41 developers from inspecting dispatch logic, adapting billing structures, or extending the software
42 to novel or emerging technologies. This is a fundamental barrier: as hybrid systems increasingly
43 incorporate electric vehicles (EVs), heat pumps, thermal batteries, peer-to-peer net metering,
44 and novel tariff structures, the inability to modify or audit HOMER Pro's internal models has
45 become a critical limitation for rigorous research and reproducible science.

46 Open-source alternatives exist but remain narrowly scoped. Most PV simulation tools, including
47 SAM ([National Renewable Energy Laboratory, 2023b](#)), PVWatts ([National Renewable Energy
48 Laboratory, 2023a](#)), PVsyst ([PVsyst SA, 2023](#)), and RETScreen ([Natural Resources Canada,
49 2023](#)), provide performance simulation or financial analysis for individual technologies but do
50 not perform multi-component hybrid system optimization. None implement the complex utility
51 tariff structures needed for real-world economic analysis of grid-interactive systems, and none
52 support EVs or heat pumps. SAMA was developed specifically to fill this gap: providing a fully
53 open, AGPL-3.0-licensed, validated alternative to HOMER Pro that extends its capabilities to
54 emerging load types, system components and tariff regimes while remaining freely accessible
55 to the global research community ([Sadat et al., 2023](#); [Sadat & Pearce, 2025](#)).

56 State of the Field

57 A comprehensive survey of available PV and hybrid energy system software tools was conducted
58 by Sadat et al. ([Sadat et al., 2023](#)), covering different tools across simulation, design,
59 and optimization categories. The findings are summarized here to contextualize SAMA's
60 contribution.

61 Tools such as the NREL System Advisor Model (SAM) ([National Renewable Energy Laboratory,
62 2023b](#)) enable exploration of the financial and physical parameters of solar and storage systems,
63 but focus on single-technology performance simulation rather than multi-component hybrid
64 optimization. PVWatts ([National Renewable Energy Laboratory, 2023a](#)), developed by NREL,
65 estimates energy production and cost savings for grid-connected PV systems and is widely
66 used for preliminary evaluations, but does not support hybrid system sizing. RETScreen
67 (Natural Resources Canada) ([Natural Resources Canada, 2023](#)) assists in energy generation
68 and conservation decisions across diverse project types, including hybrid wind-PV-generator
69 configurations, but does not perform optimal sizing of hybrid energy systems. PVsyst ([PVsyst
70 SA, 2023](#)) provides detailed simulation of grid-connected, stand-alone, pumping, and DC-
71 grid PV systems with a comprehensive meteorological and component database, but again
72 without optimization capability for hybrid multi-source configurations. TRNSYS ([Solar Energy
73 Laboratory, University of Wisconsin–Madison, 2023](#)), a modular transient systems simulation
74 environment developed at the University of Wisconsin, is highly flexible for dynamic energy
75 system analysis but requires significant user expertise and does not include built-in HES
76 optimization.

77 As demonstrated in the comparative table of Sadat et al. ([Sadat et al., 2023](#)), HOMER Pro
78 ([HOMER Energy, 2024b](#)) and HOMER Grid ([HOMER Energy, 2024a](#)) are the only tools in the
79 field that provide optimal sizing of hybrid energy systems. HOMER Pro, originally developed
80 as open-source software by NREL, offers optimization of PV, wind, battery, diesel generator,
81 grid, hydrogen, hydro, and biomass combinations with cost sensitivity and emissions analysis.
82 HOMER Grid is designed to optimize the value of behind-the-meter distributed energy systems
83 for commercial grid-connected applications. Both remain the dominant commercial tools in
84 the field; however, they are closed-source, costly, and cannot be adapted or extended by users,
85 a fundamental limitation for research reproducibility and for modeling emerging technologies.

86 SAMA's key differentiators within this landscape are: (1) full source transparency under AGPL-
87 3.0 licensing, enabling users to inspect, adapt, and extend all dispatch logic, tariff structures,
88 and component models, which is a fundamental requirement for reproducible research that no
89 commercial alternative provides ([Sadat, 2023](#)); (2) comprehensive dynamic tariff and lifecycle
90 economic modeling covering various electricity and natural gas rate structures, including TOU,

91 tiered, seasonal, and ultra-low TOU regimes, with annual price escalation, adjustments and
92 taxations, and renewable incentive modeling (Sadat & Pearce, 2025); (3) a novel EV energy
93 management system with departure-aware scheduling, three-tier V2G arbitrage, renewable
94 surplus prioritization, and adaptive safety buffering; (4) integrated heat pump performance
95 models using manufacturer lookup data for hourly COP estimation; (5) user-configurable system
96 topology, allowing optimization of any combination of PV, wind, battery, diesel generator,
97 EV, and heat pump components; and (6) four interchangeable metaheuristic optimizers (PSO,
98 ADE, ABC, GWO) enabling algorithmic comparison within a consistent evaluation framework.

99 Software Design

100 SAMA is structured as a modular Python package with six subsystems: core input configuration,
101 component physics models, an hourly energy management system (EMS), tariff and billing
102 calculators, metaheuristic optimizers, and result reporting. Key design decisions included
103 prioritizing numerical performance for the inner dispatch loop (addressed through Numba JIT
104 compilation), separating tariff logic into independent calculator modules to accommodate
105 diverse utility structures, and using a singleton input configuration object to allow external
106 YAML-based configuration without modifying source code. Full technical documentation is
107 provided in the repository (Sadat, 2023).

108 **Component models.** The solar PV model applies a temperature-dependent efficiency formulation
109 with nominal operating cell temperature (NOCT) correction using plane-of-array irradiance
110 from the National Solar Radiation Database (NSRDB) (National Renewable Energy Laboratory,
111 2022). The wind turbine model applies a power-law wind shear profile and a piecewise power
112 curve. Battery storage supports lithium-ion (constant-efficiency model with lifetime throughput
113 degradation) and lead-acid chemistries via the Kinetic Battery Model (KiBaM) (Manwell
114 & McGowan, 1993). The diesel generator uses a linear fuel consumption curve. Heat pump
115 performance is represented by manufacturer lookup tables for Bosch and Goodman systems,
116 returning hourly COP values as functions of ambient temperature and building thermal load.
117 The EV model tracks state-of-charge dynamics with round-trip efficiency.

118 **Energy management system.** The base EMS (EMS.py) is Numba JIT-compiled for performance.
119 In each of annual hours, the dispatcher resolves net load using marginal-cost-based prioritization
120 across the battery, diesel generator, and utility grid. Six dispatch scenarios are defined from the
121 relative values of the grid buy price, diesel cost, and battery wear cost, enabling cost-optimal
122 decisions at each time step. Unmet energy accumulates toward the loss of power supply
123 probability (LPSP) constraint.

124 The EV extension (EMS_EV.py) manages V2H and V2G dispatch through a departure-aware
125 scheduler that dynamically identifies the next departure time, prioritizes renewable surplus for
126 required charging, and opportunistically pre-charges from the grid during low-price windows.
127 For V2G, a three-tier arbitrage engine selects between aggressive, moderate, and conservative
128 discharge strategies based on price percentile thresholds, profit margin requirements, time-
129 to-departure flexibility, and inverter capacity. A rolling safety buffer derived from historical
130 departure data ensures the EV always meets its required departure SOC, and all discharge
131 decisions undergo final net-profit validation accounting for battery wear cost, round-trip
132 efficiency losses, and solar displacement penalties. The heat pump post-processor (EMS_HP.py)
133 attributes hourly electricity consumption to heating versus cooling loads using proportional
134 and marginal-cost-based methods.

135 **Tariff and financial modeling.** SAMA implements eight electricity rate structures and eight
136 natural gas rate structures, covering flat, seasonal, monthly, tiered, TOU, and ultra-low
137 TOU tariffs used across North American utilities (Sadat & Pearce, 2025). Net metering is
138 modeled with annual credit reconciliation. The lifecycle financial projection computes NPC
139 using the capital recovery factor (CRF) with real and nominal discount rates, applies annual
140 price escalation for grid electricity and natural gas, schedules component replacements, and

141 applies renewable energy incentives (such as Investment Tax Credits (ITCs)). PV system cost
142 benchmarks follow NREL methodology (Ramasamy et al., 2022).

143 **Optimization framework.** The five design variables (PV capacity, wind turbine count, battery
144 count, diesel generator size, and inverter rating) are explored by any of the four included
145 metaheuristic algorithms. ADE uses different mutation strategies with adaptive scaling and
146 crossover parameters (Das & Suganthan, 2011; Storn & Price, 1997). ABC uses multi-
147 dimensional perturbations with guided scout reinitialization (Karaboga & Basturk, 2007).
148 GWO implements social hierarchy convergence with an optional parallel multiprocessing variant
149 (Mirjalili et al., 2014). PSO uses inertia-weight damping with cognitive and social learning
150 coefficients (Kennedy & Eberhart, 1995; Shi & Eberhart, 1998). All four algorithms share the
151 same fitness function and smooth constraint penalty framework, enabling direct performance
152 comparisons.

153 **Interfaces and access.** SAMA can be accessed in three ways. First, it is installable via pip
154 install samapy and provides two command-line entry points: samapy-config (an interactive
155 configuration wizard) and samapy-run (an optimization runner accepting YAML configuration
156 files). Second, users can download or clone the complete source code directly from the GitHub
157 repository at <https://github.com/Sas1997/SAMA> (Sadat, 2023) and execute SAMA using
158 any Python-compatible IDE such as PyCharm, or from the command line in any Python 3.9+
159 environment. Third, a graphical user interface (GUI) based Windows executable (.exe) currently
160 on alpha development stage is available for users who prefer not to install a Python environment;
161 this is distributed on GitHub (Sadat, 2026). A Python API also enables programmatic access for
162 integration with external workflows. Comprehensive user documentation, including installation
163 instructions, input file specifications, and worked examples, is provided in the repository
164 documentation (Sadat, 2023).

165 Research Impact Statement

166 SAMA has supported six peer-reviewed publications. The primary paper describes the core
167 platform and validation against HOMER Pro across Sacramento, California and New Bern,
168 North Carolina, demonstrating close agreement in optimal system configurations, LCOE, NPC,
169 and battery state-of-charge profiles (Sadat et al., 2023). Subsequent studies applied SAMA
170 to analyze economic grid defection thresholds for U.S. households (Sadat & Pearce, 2024),
171 evaluate the impact of electricity pricing structures on PV and PV-battery systems across
172 Canadian utilities (Sadat & Pearce, 2025), quantify PV systems as inflation hedges (Sadat et
173 al., 2025), model peer-to-peer virtual net metering schemes (Groza et al., 2024), and analyze
174 subsidy redirection scenarios in Alberta (Sadat et al., 2026). The package is indexed on PyPI
175 and the source repository is publicly accessible on GitHub (Sadat, 2023).

176 AI Usage Disclosure

177 Generative AI tools (Claude Sonnet 4.5, Anthropic, 2025-2026) were used to assist with editing
178 and restructuring the user documentation, converting documentation from Word to Markdown
179 format, and editing portions of this paper's prose for clarity. Human authors reviewed, edited,
180 and validated all AI-assisted outputs for technical accuracy, originality, and compliance with
181 JOSS requirements. The authors bear full responsibility for the accuracy, originality, and ethical
182 compliance of all submitted materials.

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